common—or we might be on completely the wrong track in believing that singularities are made of quantum foam.

Clear answers might come in the next decade or two from research now being carried out by Stephen Hawking, James Hartle, and others, building on foundations laid by John Wheeler and Bryce DeWitt.[3]

Most everything in the Universe changes with age: Stars consume their fuel and die; the Earth gradually loses its atmosphere by evaporation into space and ultimately will become an airless, dead planet; and we humans grow wrinkled and wise.

The tidal forces deep inside a black hole, near its singularity, are no exception. They, too, must change with age, according to calculations done in 1991 by Werner Israel and Eric Poisson of the University of Alberta, and Amos Ori, a postdoc in my Caltech group (building on earlier work of Andrei Doroshkevich and Igor Novikov). When the hole is newborn, its interior tidal forces exhibit violent, chaotic, BKL-type oscillations (Figure 13.6 above). However, as the hole ages, the chaotic oscillations become tamer and gentler, and gradually disappear.

For example, an astronaut who falls into a 10-billion-solar-mass hole in the core of a quasar within the first few hours after the hole is born will be torn apart by wildly oscillating BKL tidal forces. However, a second astronaut, who waits until a day or two after the hole is born before plunging inside, will encounter much more gently oscillating tidal forces. The tidal stretch and squeeze are still large enough to kill the second astronaut, but being more gentle than the day before, the oscillating stretch and squeeze will allow the second astronaut to survive longer, and approach closer to the singularity before he dies, than did the first astronaut. A third astronaut, who waits until the hole is many years old before taking the plunge, will face an even gentler fate. The tidal forces surrounding the singularity have now become so tame and meek, according to Israel's, Poisson's, and Ori's calculations, that the astronaut will hardly feel them at all. He will survive, almost unscathed, right up to the edge of the probabilistic quantum gravity singularity. Only at the singularity's edge, just as he comes face-to-face with the laws of quantum gravity, will the astronaut be killed—and we cannot even be absolutely sure he gets killed